QUANTS INTERMEDIATE - 5

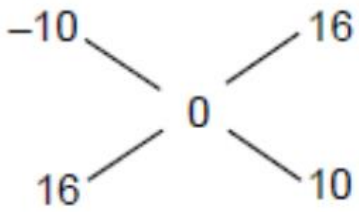
Directions for questions 1 and 2: Select the correct alternative from the given choices.

Anish sold two articles gaining 16% on one and incurring a loss of 10% on the other. What is the cost price of the first article, if he neither gained nor lost in the overall transaction, and the total cost price of the articles is Rs.78,000?

- a) Rs.48,000
- b) Rs.30,000
- c) Rs.40,000
- d) Rs.38,000

Solution:

Let the cost price of the first article be ₹x.
Using alligation



$$\frac{16}{10} = \frac{78000 - x}{x}$$

$\Rightarrow x = ₹30,000.$ Choice (B)

Directions for questions 1 and 2: Select the correct alternative from the given choices.

K =

$$\frac{x^{2(a+b)}+6x^{a+b}+9}{x^{3a-5b}(x^{a+2b}+3x^{b})(x^{a+2b}+3x^{b})}$$

where x is not equal to 0. The value of K is _____.

- a) $\frac{x^{3b}}{x^{3a}}$
- b) $\frac{x^{3a}}{x^{3b}}$
- c) 1
- d) 2

Solution:

$$K=\frac{\left(x^{2(a+b)}+6x^{a+b}+9\right)}{x^{3a-5b}\left(x^{a+2b}+3x^{b}\right)\left(x^{a+2b}+3x^{b}\right)}$$

$$=\frac{\left(x^{a+b}+3\right)^{2}}{x^{3a-5b}\left(x^{a+b}+3\right)x^{b}\left(x^{a+b}+3\right)}$$

$$\frac{(x^{a+b}+3)^{2}}{x^{3a-5b}x^{2b}(x^{a+b}+3)^{2}}=\frac{1}{x^{3a-3b}}=\frac{x^{3b}}{x^{3a}}$$

Choice (A)

Directions for questions 3 and 4: Type in your answer in the input box provided below the question.

The sum of the squares of the digits of a three-digit number is equal to S. Each of its digits is multiplied by the sum of its other two digits. The sum of all such possible results is denoted by R. If S = R/2, how many such three-digit numbers exist?

Solution:

Let the three-digit number satisfying the given conditions be denoted by abc.

$S = a^2 + b^2 + c^2$

$R = a(b + c) + b(a + c) + c(a + b) = 2(ab + bc + ca)$

Given $a^2 + b^2 + c^2 = \frac{1}{2}(2(ab + bc + ca))$

$\Rightarrow \frac{1}{2}[(a - b)^2 + (b - c)^2 + (c - a)^2] = 0$

The above equation would be satisfied only if

$a - b = b - c = c - a = 0$

i.e., $a = b = c$.

$\therefore$ abc has 9 possibilities. Ans : (9)

Directions for questions 3 and 4: Type in your answer in the input box provided below the question.

A test has 90 questions. Each correct answer fetches 1 mark, each wrong answer carries a penalty of

$\frac{1}{3}^{rd}$

mark, each unanswered question carries a penalty of

$\frac{1}{6}^{th}$

the mark. A student who attempted the test scored 51 marks. The number of questions answered correctly by the student is at least

Solution:

Let the number of questions correctly answered, wrongly answered and unanswered by the student be C, W and U respectively.

$C + W + U = 90$ and $C - \frac{1}{3}W - \frac{1}{6}U = 51$ i.e.,

$6C - 2W - U = 306$

Adding these, $7C - W = 396$

$C = \frac{396 + W}{7}$

C is an integer. $\therefore 396 + W$ must be divisible by 7

$\therefore$ Least (W) is 3

As least (W) is 3, least (C) is 57

The student answered at least 57 questions correctly. Ans : (57)

Directions for questions 3 and 4: Type in your answer in the input box provided below the question.

A test has 90 questions. Each correct answer fetches 1 mark, each wrong answer carries a penalty of

$\frac{1}{3}^{rd}$

mark, each unanswered question carries a penalty of

$\frac{1}{6}^{th}$

the mark. A student who attempted the test scored 51 marks. The number of questions answered correctly by the student is at least

Solution:

Let the number of questions correctly answered, wrongly answered and unanswered by the student be C, W and U respectively.

$C + W + U = 90$ and $C - \frac{1}{3}W - \frac{1}{6}U = 51$ i.e.,

$6C - 2W - U = 306$

Adding these, $7C - W = 396$

$C = \frac{396 + W}{7}$

C is an integer. $\therefore$ 396 + W must be divisible by 7

$\therefore$ Least (W) is 3

As least (W) is 3, least (C) is 57

The student answered at least 57 questions correctly. Ans : (57)

Directions for questions 5 to 7: Select the correct alternative from the given choices.

PQRS is a cyclic quadrilateral. PR and QS meet at O. If

$\angle$

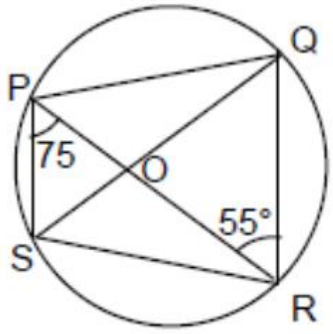
SPR = 75° and

$\angle$

PRQ = 55°, what is the measure of $\angle POQ$?

- a) 130°
- b) 110°
- c) 120°
- d) 140°

Solution:



$\angle PSQ = \angle PRQ = 55°$
(Angles subtended by the same segment)
$\angle POS = 180 - (75 + 55) = 50°$
$\angle POQ = 130°$ Choice (A)

Directions for questions 5 to 7: Select the correct alternative from the given choices.

a, b and c are positive numbers not equal to 1. If $\log_a 100$, $2\log_b 10$, $(2\log_c 5 + \log_c 4)$ are in harmonic progression, then a, b, c are in _____.

- a) arithmetic progression.
- b) geometric progression.
- c) harmonic progression.
- d) Both (A) and (B)

Solution:

Given $\log_a 100$, $2\log_b 10$, $2\log_c 5 + \log_c 4$ are in H.P.

$\Rightarrow \frac{\log 100}{\log a}, \frac{\log 100}{\log b}, \frac{\log 100}{\log c}$, are in H.P.

$\Rightarrow$ log a, log b, log c are in A.P.

$\Rightarrow 2 \log b = \log a + \log c \Rightarrow b^2 = ac$

$\therefore$ a, b, c are in G.P. Choice (B)

Directions for questions 5 to 7: Select the correct alternative from the given choices.

If $|2x^2 - 9x + 9| > 2x^2 - 9x + 9$, the range of x is _____.

- a) $\left(\frac{1}{2},3\right)$
- b) $\left(\frac{3}{2},5\right)$
- c) $\left(\frac{3}{2},3\right)$
- d) $\left(\frac{1}{2},5\right)$

Solution:

Let $y = 2x^2 - 9x + 9$

$|y| > y$ (given)

This is only possible when y is negative

$2x^2 - 9x + 9 < 0$

$(2x - 3)(x - 3) < 0$

$\frac{3}{2} < x < 3.$ Choice (C)

Directions for question 8: Type in your answer in the input box provided below the question.

If p = $\sqrt{5} - 2$, find the value of $p^4 + 8p^3 + 16p^2 + 4$.

Solution:

Given $p = \sqrt{5} - 2 \Rightarrow p + 2 = \sqrt{5}$

$\Rightarrow p^2 + 4p + 4 = 5 \Rightarrow p^2 + 4p = 1$

$\Rightarrow p^4 + 16p^2 + 8p^3 = 1$

$\Rightarrow p^4 + 8p^3 + 16p^2 + 4 = 5$ Ans : (5)

Directions for questions 9 and 10: Select the correct alternative from the given choices.

Seven friends, including Ravi, decided to donate some money to a charity. Each of the six friends other than Ravi donated Rs.90 and Ravi donated Rs.60 more than the average contribution of the group of seven. How much did Ravi donate?

- a) Rs.175
- b) Rs.160
- c) Rs.150
- d) Rs.100

Solution:

Let the amount donated by Ravi be ₹x

$$\frac{6 \times 90 + x}{7} + 60 = x$$

$960 + x = 7x$

$\Rightarrow x = 160.$ Choice (B)

Directions for questions 9 and 10: Select the correct alternative from the given choices.

How many real solutions does the equation $x^6+19x^3=216$ have?

- a) 1
- b) 2 (Correct Answer - ✓)
- c) 4
- d) 6

Solution:

$x^6 + 19x^3 - 216 = 0$
$\Rightarrow (x^3+27)(x^3-8) = 0 \Rightarrow x = -3$ or 2.
The other 4 roots are imaginary. There are only 2 real roots. Choice (B)

Directions for question 11: Type in your answer in the input box provided below the question.

The maximum distance between any point on $(x + 5)^2 + y^2 = 16$ and any point on $x^2 + (y + 12)^2 = 36$ is

Solution:

Given circles are
$(x + 5)^2 + y^2 = 16$ ..... (1)
$x^2 + (y + 12)^2 = 36$ ..... (2)
The centre and the radius of the first circle are C_1 (–5,0) and $r_1 = 4$, while the centre and the radius of the second circle are C_2(0, –12) $r_2 = 6$.
The distance between two centres is C_1C_2 (d) = $\sqrt{(-5)^2 + (-12)^2} = 13$
$r_1 + r_2 = 4 + 6 = 10$.
As $d > r_1 + r_2$, the two circles do not intersect. Let P be any point on (1) and Q be any point on (2). The distance between P and Q is maximum when line PQ passes through the centres of the two circles.
$\therefore PQ = PC_1 + C_1C_2 + C_2Q$
C_1P is radius of (1) circle
C_2Q is radius of (2) circle
$= 4 + 13 + 6 = 23$.
Ans : (23)

Directions for questions 12 to 15: Select the correct alternative from the given choices.

At how many points does the line $4x - y = 4$ intersect the curve $y = x^2$?

- a) 9
- b) 2
- c) 1
- d) None of these

Solution:

The point or points where the live $4x - y = 4$ intersects the curve $y = x^2$ are obtained by solving the two equations
$\therefore x^2 - 4x + 4 = 0$ or $x = 2$.
The only point of intersection is (2, 4).
Choice (C)

Directions for questions 12 to 15: Select the correct alternative from the given choices.

The roots of $x^2 + (8m + 8)x + 32m = 0$ are _____.

- a) real and distinct
- b) real and equal
- c) real
- d) non-real

Solution:

Discriminant of the equation (D) = $(8m + 8)^2 - 4(32m)$
$= 64(m^2 + 1)$.
D will be positive. [∵ Least D = 64]
∴ The roots are real and distinct. Choice (A)

Directions for questions 12 to 15: Select the correct alternative from the given choices.

The sum of the factors of 44444444 is _____.

- a) 86385828
- b) 87496416 (Correct Answer - ✓)
- c) 88342952
- d) 89483276

Solution:

$$44444444 = 4(11111111) = \frac{4(10^8 - 1)}{9}$$

$$= \frac{4}{9}(10^4 - 1)(10^4 + 1) = 4(1111)(10001)$$

$$= 4(1111)(105^2 - 32^2) = 2^2(11)(101)(73)(137)$$

The sum of all the factors of 44444444

$$= \left(\frac{2^3 - 1}{2 - 1}\right)\left(\frac{11^2 - 1}{11 - 1}\right)\left(\frac{101^2 - 1}{101 - 1}\right)\left(\frac{73^2 - 1}{73 - 1}\right)\left(\frac{137^2 - 1}{137 - 1}\right)$$

$$= 7\,(12)\,(102)\,(74)\,(138) = 87496416$$

Choice (B)

Directions for questions 12 to 15: Select the correct alternative from the given choices.

A committee of four is to be formed from eight persons P_1 through P_8. P_1 and P_2 are best friends and each of them wants to serve in the committee only if the other also serves in the committee. In how many ways can the committee he formed?

- a) 20
- b) 30
- c) 25
- d) 35

Solution:

P_1 and P_2 want to serve in the committee together or not at all.
If P_1 and P_2 serve in the committee together, number of ways in which the remaining persons can be chosen = ${}^6C_2 = 15$. The committee can be formed in ${}^6C_2$ i.e. 15 ways.
If P_1 and P_2 do not serve in the committee the four persons who have to serve in the committee can be any four of the remaining six persons. The committee can be formed ${}^6C_4$ i.e. 15 ways.
The committee can be formed in 30 ways.

Choice (B)